

Michelle Dominic

✉ mmdomini@uwaterloo.ca | in michelle-dominic | 🌐 mmdominic | 📧 m76domi98

TECHNICAL SKILLS

Languages: C, C++, Python, SystemVerilog, VHDL, JavaScript, TypeScript, C#, Java, SQL
Embedded & Hardware: QNX, STM32, ESP32, Raspberry Pi, FPGA Design, UART, SPI, I2C, PWM, KiCad
Frameworks & Libraries: TensorFlow, Scikit-learn, XGBoost, Hugging Face, React, Vue.js, Node.js, FastAPI
Tools & Platforms: Git, Docker, GitHub Actions, Quartus Prime, ModelSim, Azure OpenAI, Azure Application Insights, PostgreSQL, MongoDB, Power BI, Postman
Certifications: HackerRank SQL, Scalar React JS

EDUCATION

University of Waterloo Waterloo, ON, Canada
Candidate for BAsC in Computer Engineering 2024–2029

EXPERIENCE

FPGA Design Member Jul. 2026 – Present
UWASIC Waterloo, ON

- Developing **SystemVerilog RTL** for IPv4 header extraction in a **10 Gbps Ethernet parser**
- Building verification testbenches to validate packet parsing, metadata generation, and protocol correctness
- Collaborating through Git, code reviews, simulation, and iterative RTL development

Product Management Intern Sept. 2025 – Dec. 2025 & May 2026 – Aug. 2026
SAP Waterloo, ON

- Defined MVP requirements for a unified **SAP HANA Cloud** application to manage cloud instances
- Queried **Azure Application Insights** telemetry and used **Azure OpenAI embeddings** to cluster customer interactions, identify feature requests, and prioritize AI agent opportunities
- Built **Power BI** dashboards tracking HANA Cloud Central feature adoption, supporting **30+** Product Managers
- Validated partner MVP integrations and authored partnership documentation, testing guides, & technical preview
- Led a customer preview initiative by recruiting users and deploying MVP builds using **AWS S3**, & **CloudFront**

Full Stack Developer Intern Jan. 2025 – Apr. 2025
Stubbe's Precast Harley, ON

- Built Python and SQL data pipelines, reducing preparation time by **35%**
- Developed forecasting models using **TensorFlow**, **XGBoost**, and **Scikit-learn**, reducing prediction error by **15%**
- Integrated an AI chatbot into internal manufacturing software using **C#**, **.NET**, **Node.js**, and **Vue.js**

PROJECTS

hand.shake: AI Grip-Assist Glove 📧 | QNX, Raspberry Pi 5, ESP32, Python, Gemini API Jul. 2026

- Built an AI-powered assistive glove combining a **QNX-powered Raspberry Pi 5, ESP32**, computer vision, and tendon-driven actuation to improve grip strength
- Developed a computer vision pipeline on **QNX 8** that captured images, identified objects using **Gemini** with fallback vision-language models, and selected appropriate grip assistance
- Programmed **ESP32 firmware** for PWM servo control, wireless communication, and real-time telemetry to monitor assistive force during operation

MeshGit 📧 | React, Three.js, Node.js, Python, PostgreSQL, Docker 2026

- Built a geometry-aware version control system for STL mesh files with interactive Three.js visual diffs
- Designed a microservice architecture with a Node.js API and Python geometry processing service
- Implemented commit history, branching, conflict detection, GitHub integration, and Docker-based deployment

Summus: AI Terms & Conditions Assistant 📧 | JavaScript, FastAPI, Python, Ollama Apr. 2025

- Built a Chrome extension that summarizes terms & conditions and explains legal documents using AI
- Developed a FastAPI backend integrating Ollama and Hugging Face models through REST APIs
- Implemented the frontend with JavaScript, HTML/CSS, and validated APIs using Postman

Home Energy Monitor & HVAC Controller | VHDL, Quartus Prime, ModelSim, Intel MAX 10 FPGAMay 2025

- Designed an FPGA-based HVAC controller using synchronous VHDL on an Intel MAX 10 FPGA
- Verified modular components in ModelSim and integrated the complete design in Quartus Prime
- Displayed system state and temperature on dual seven-segment displays for real-time hardware feedback